

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12411027
Kind Code	B2
Date of Patent	September 09, 2025
Inventor(s)	Douseki; Takakuni et al.

Power generation sensor

Abstract

A power generation sensor includes: a housing body in which a module receiving a generated power is housed; a module with a first connector electrode disposed on a front face; a second connector electrode disposed on an inner face of the housing body connected to the first connector electrode; and a power generation electrode connected to the second connector electrode.

Inventors: Douseki; Takakuni (Shiga, JP), Tanaka; Ami (Shiga, JP), Imoto; Shiyuya (Shiga, JP), Utsunomiya; Fumiyasu (Nagano, JP)

Applicant: The Ritsumeikan Trust (Kyoto, JP); ABLIC Inc. (Nagano, JP)

Family ID: 89261070

Assignee: The Ritsumeikan Trust (Kyoto, JP); ABLIC Inc. (Nagano, JP)

Appl. No.: 18/324161

Filed: May 26, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240003718 A1	Jan. 04, 2024

Foreign Application Priority Data

JP	2022-104631	Jun. 29, 2022
----	-------------	---------------

Publication Classification

Int. Cl.: A61F13/42 (20060101); G01D11/24 (20060101); A61F13/49 (20060101); G01N33/487 (20060101); G01N33/493 (20060101)

U.S. Cl.:

CPC **G01D11/245** (20130101); A61F13/42 (20130101); A61F13/49 (20130101);
G01N33/48785 (20130101); G01N33/493 (20130101)

Field of Classification Search

CPC: G01D (11/245); G01N (33/493); A61F (13/42); A61F (13/49)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2024/0003718	12/2023	Douseki	N/A	G01D 11/245
2024/0004099	12/2023	Douseki	N/A	G01V 3/088

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2015229003	12/2014	JP	N/A
2017032331	12/2016	JP	N/A
WO-2013061963	12/2012	WO	A61F 13/42

Primary Examiner: Williams; Jamel E

Attorney, Agent or Firm: JCIPRNET

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application claims the priority benefits of Japanese application no. 2022-104631, filed on Jun. 29, 2022. The entirety of each of the above-mentioned patent applications is hereby incorporated by reference herein and made a part of this specification.

BACKGROUND

Technical Field

(2) The present invention relates to a power generation sensor.

Description of Related Art

(3) Japanese Patent Laid-Open No. 2015-229003 discloses a power generation module including an electrode generating an electromotive force in accordance with contact with urine. Such a power generation module can be used as a power generation sensor.

(4) In a power generation sensor, there is a request for easy connection of a module receiving generated electric power to a power generation electrode. For example, a case in which a module is connected to a power generation electrode mounted in a member such as a diaper attached to a human body is assumed.

SUMMARY

(5) The present invention provides a power generation sensor capable of causing easy connection of a module receiving generated electric power to a power generation electrode.

(6) According to an embodiment of the present invention, a power generation sensor includes a housing body in which a module receiving generated electric power is housed, a module having a front face on which a first connector electrode is disposed, a second connector electrode connected to the first connector electrode disposed on a housing body inner face, and a power generation

electrode connected to the second connector electrode.

(7) According to the present invention, a module can be easily connected to a power generation electrode.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a schematic configuration diagram of a sensor according to an embodiment.

(2) FIG. 2 is a diagram illustrating a method of manufacturing a sensor.

(3) FIG. 3 is a diagram illustrating a method of connecting a module to a sensor.

(4) FIG. 4 is a diagram illustrating another example of a method of connecting a module to a sensor.

(5) FIG. 5 is a diagram illustrating an example of a contact part.

(6) FIG. 6 is a schematic diagram illustrating a configuration of a module.

(7) FIG. 7 is a schematic diagram illustrating an overview of a case in which a sensor is used in a detection system.

(8) FIG. 8 is a cross-sectional view of an absorption member main body along line VIII-VIII illustrated in FIG. 7.

(9) FIG. 9 is a cross-sectional view of an absorption member main body along line IX-IX illustrated in FIG. 7.

(10) FIG. 10 is a diagram illustrating results of measurements of wireless signals acquired through the inventors' tests of a detection system.

DESCRIPTION OF THE EMBODIMENTS

(11) Further details will be described in the following embodiment.

(12) (1) According to an embodiment, there is provided a power generation sensor including: a housing body in which a module receiving generated power is housed; a module with a first connector electrode disposed on a front face; a second connector electrode disposed on an inner face of the housing body; and a power generation electrode further connected to the second connector electrode, in which the first connector electrode and the second connector electrode are connected to each other.

(13) Since the housing body in which the module is housed is included, and the second connector electrode is disposed on the inner face thereof, by housing the module with the first connector electrode disposed on the front face thereof in the housing body, the first connector electrode is brought into contact with the second connector electrode. In accordance with this, positioning performed at the time of connecting the first connector electrode and the second connector electrode becomes easy, and the module can be easily mounted. As a result, the module can be easily connected to the power generation electrode. In addition, the exposure of the module is inhibited in accordance with housing of the module in the housing body, and damage and contamination of the module can be inhibited.

(14) (2) The power generation sensor of (1), preferably, further including a contact part bringing the first connector electrode of the module housed in the housing body and the second connector electrode into contact with each other. In accordance with this, the first connector electrode and the second connector electrode can be easily brought into contact with each other, and the contact state can be easily maintained. As a result, connection between the first connector electrode and the second connector electrode can be easily maintained.

(15) (3) In the power generation sensor of (2), it is preferable that the contact part include a pressing member pressing the module housed in the housing body to the inner face of the housing body. In accordance with this, the module is pressed to the inside of the housing body. As a result, the first connector electrode and the second connector electrode can be easily connected to each

other, and the connection is maintained.

(16) (4) In the power generation sensor of (1), it is preferable that the housing body include an insertion part for the module. In accordance with this, the module can be inserted into the housing body.

(17) (5) In the power generation sensor of (1), it is preferable that a positive electrode of the first connector electrode be disposed on a first front face of the module, that a negative electrode of the first connector electrode be disposed on face opposite the first front face, that a positive electrode of the second connector electrode be disposed on a first face of the inner face of the housing body, that a negative electrode of the second connector electrode be disposed on a second face positioned in a first direction with respect to the first face, that the power generation electrode include a positive electrode connected to the positive electrode of the second connector electrode and a negative electrode connected to the negative electrode of the second connector electrode, and that the positive electrode and the negative electrode of the power generation electrode be disposed apart from each other in a second direction different from the first direction. In accordance with this, the positive electrode and the negative electrode of the power generation electrode are connected to the module respectively through the second connector electrode and the first connector electrode.

(18) FIG. 1 is a schematic configuration diagram of a sensor 1 according to this embodiment. FIG. 2 is a diagram illustrating a method of manufacturing the sensor 1. FIG. 3 is a diagram illustrating a method of connecting a module 3 to the sensor 1. FIG. 4 is a diagram illustrating another example of a method of connecting the module 3 to the sensor 1. FIG. 5 is a diagram illustrating an example of a contact part.

(19) The sensor 1 includes a first sensor member 1A and a second sensor member 1B and is configured by overlapping those members. An electrode 11 is disposed in the first sensor member 1A, and an electrode 12 is disposed in the second sensor member 1B. The first sensor member 1A includes a base 13A used for disposing the electrode 11, and the second sensor member 1B includes a base 13B used for disposing the electrode 12.

(20) For example, the bases 13A and 13B are composed of a liquid absorption material. For example, the bases 13A and 13B are fabrics and, for example, are formed of fabrics of a cotton material. The bases 13A and 13B have the same shape or approximately the same shape.

(21) Each of the bases 13A and 13B includes a first base part 131 used for configuring a housing body 16 to be described below and a second base part 132 in which electrodes 11 and 12 are disposed. The second base part 132 is long in one direction and has one end at which the first base part 131 is disposed. With respect to the sensor 1, a length direction of the second base part 132 will be defined as a y axis, and a lateral direction orthogonal to the length direction will be defined as an x axis. The first base part 131 is disposed at an end of the second base part 132 in a positive direction of the y axis.

(22) One pair of electrodes 11 and 12 are electrodes for power generation. The electrode 11 functions as a positive electrode, and the electrode 12 functions as a negative electrode. The electrodes 11 and 12 generate electric power by being brought into contact with a liquid present between the electrodes. In accordance with this, the sensor 1 functions as a power generation sensor and can detect presence of a liquid between the electrodes 11 and 12 in accordance with generated electric power of the electrodes 11 and 12.

(23) The electrodes 11 and 12 are composed of different materials. Each of the electrodes 11 and 12 has a structure of a thread shape or a band shape. As one example, the electrode 11 is composed of a silver thread, and the electrode 12 is composed of an aluminum thread. By configuring the electrodes 11 and 12 using threads, the electrodes 11 and 12 can be easily formed using sewing to be described below. In addition, widths W4 and W5 of the electrodes 11 and 12 in the x-axis direction can be narrowed, and a decrease in the strength can be inhibited. Any material may be used for the electrode 11 as long as it has a structure of a thread shape or a band shape and can

serve as a positive electrode, and for example, magnesium, zinc, or iron may be used. Any material may be used for the electrode **12** as long as it has a structure of a thread shape or a band shape and can serve as a negative electrode, and, for example, tin, lead, copper, stainless steel, carbon, gold, or platinum may be used.

(24) The electrodes **11** and **12** are respectively woven into the second base parts **132** of the bases **13A** and **13B** which are fabrics in the y-axis direction. More specifically, in this example, both of the electrodes **11** and **12** have a thread-shape structure having conductivity, the electrode **11** is woven into the second base part **132** of the base **13A**, and the electrode **12** is woven into the second base part **132** of the base **13B**.

(25) Preferably at least one of the electrodes **11** and **12** has a surface area of a negative side in the y-axis direction, that is, a rear side, which is larger than a surface area of a positive side in the y-axis direction, that is, a front side. The surface areas represent areas in contact with an absorption body **53** when mounted in an absorption member main body **5** to be described below and represent surface areas of parts of the electrodes **11** and **12** formed using threads, woven into the bases **13A** and **13B** which are exposed on at least front faces of the bases **13A** and **13B** on the side of the absorption body **53**.

(26) As one example, a first part **111** of the electrode **11** which is a positive side in the y-axis direction, that is, the front side, and a second part **112** which is a negative side in the y-axis direction, that is, the rear side, have different stitch structures of threads with respect to the base **13A**. The stitch structure represents a structure formed by a thread with respect to a fabric by sewing the thread into the bases **13A** and **13B**. The stitch structure being different in the first part **111** and the second part **112**, for example, represents that a method for sewing the thread forming the electrode **11** is different in the first part **111** and the second part **112**. More specifically, in the electrode **11**, the second part **112** is sewn into the base **13A** using a sewing method for implementing a higher density per unit than a density for the first part **111**.

(27) For example, a sewing method having a high density per unit is a sewing method in which an area of a thread which forms the electrode **11** appearing on the front face of the base **13A** on the absorption body **53** side is large. For example, the first part **111** is sewn into the base **13A** through wave stitching, and the second part **112** is sewn into the base **13A** through back-stitching. The back-stitching represents reverse stitching or half-reverse stitching. In accordance with this, in the second part **112**, exposure of a thread forming the electrode **11** on the front face of the base **13A** is more than exposure in the first part **111**.

(28) A length (width) **W1** of the first base part **131** of each of the bases **13A** and **13B** in the x-axis direction is larger than a width **W2** of the second base part **132**. The first base part **131** forms the housing body **16** in which a module **3** receiving generated electric power is housed. In the housing body **16**, the first base parts **131** of the first sensor member **1A** and the second sensor member **1B** overlapping each other have a space **16A** configured therebetween, and the module **3** is housed inside the space **16A**.

(29) In the first base parts **131** of the bases **13A** and **13B**, connector electrodes **14** and **15** are respectively disposed. The connector electrode **14** is connected to a front-side end of the electrode **11** which is one end in the y-axis direction, and the connector electrode **15** is connected to a front-side end of the electrode **12**.

(30) The connector electrodes **14** and **15** are respectively composed of the same materials as the electrodes **11** and **12**. In this example, the electrode **11** and the connector electrode **14** are composed of silver, and the electrode **12** and the connector electrode **15** are composed of aluminum. As one example, similar to the electrode **11**, the connector electrode **14** is composed of a silver-based material, and, similar to the electrode **12**, the connector electrode **15** is composed of an aluminum-based material. In accordance with this, the connector electrodes **14** and **15** can be respectively sewn into the bases **13A** and **13B** in a series of operations with sewing operations into the bases **13A** and **13B** of the electrodes **11** and **12**. Any material may be used for the electrode **11**

as long as it has a structure of a thread shape or a band shape and can serve as a positive electrode, and, for example, magnesium, zinc, or iron may be used. In addition, any material may be used for the electrode **12** as long as it has a structure of a thread shape or a band shape and can serve as a negative electrode, and, for example, tin, lead, copper, stainless steel, carbon, gold, or platinum may be used.

(31) A length (width) **W3** of the connector electrode **14** in the x-axis direction is larger than a length (width) **W4** of the electrode **11** in the x-axis direction. In addition, a length (width) **W3** of the connector electrode **15** in the x-axis direction is larger than a length (width) **W5** of the electrode **12** in the x-axis direction. In accordance with this, when the module **3** is housed in the housing body **16** as will be described below, positioning of the housed module **3** for connection between the connector electrodes **31** and **32** and the connector electrodes **14** and **15** can be easily performed or becomes unnecessary.

(32) Preferably, the density of threads sewed into the bases **13A** and **13B** composing the connector electrodes **14** and **15** is higher than the density of threads sewed into the bases **13A** and **13B** composing the connector electrodes **11** and **12**. In accordance with this, when the module **3** is housed in the housing body **16** as will be described below, there is a high possibility of the connector electrodes **14** and **15** coming in contact with the connector electrodes **31** and **32** of the housed module **3**. As a result, not only can positioning of the module **3** for connection between the connector electrodes **31** and **32** and the connector electrodes **14** and **15** be easily performed or unnecessary, but also contact resistance between the connector electrodes **14** and **15** and the connector electrodes **31** and **32** of the housed module **3** can be lowered.

(33) As illustrated in FIG. 2, in the sensor **1**, the electrodes **11** and **12** are respectively disposed in the bases **13A** and **13B** in the y-axis direction, and the connector electrodes **14** and **15** are formed at upper ends thereof. These are formed by sewing the bases **13A** and **13B** using threads having conductivity.

(34) Next, the first sensor member **1A** and the second sensor member **1B** are disposed to overlap each other and are sewn together in a direction (the z-axis direction) orthogonal to a plane defined by the x axis and the y axis such that the shapes of the bases **13A** and **13B** coincide or roughly coincide with each other. In the bases **13A** and **13B**, seam allowances **133** are provided in edges of sides other than an uppermost side. The first sensor member **1A** and the second sensor member **1B** are configured to overlap each other and be fixed in accordance with sewing the seam allowances **133** together.

(35) At this time, as illustrated in FIG. 2, the bases **13A** and **13B** overlap each other such that the electrodes **11** and **12** do not overlap each other. In accordance with this, the electrodes **11** and **12** are disposed by leaving a space therebetween in the x-axis direction. As a result, the sensor **1** can detect presence of a liquid between the electrodes **11** and **12** in accordance with power generation.

(36) By sewing the bases **13A** and **13B** together in the seam allowances **133**, as illustrated in FIG. 3, a space **16A** is formed between the first base parts **131** of the bases **13A** and **13B**. In accordance with this, the housing body **16** can be easily configured using the space **16A** as a space used for housing the module **3**.

(37) In accordance with no disposition of the seam allowance **133** on the uppermost sides of the bases **13A** and **13B**, when the bases **13A** and **13B** are sewed together in the seam allowances **133**, as illustrated in FIG. 3, the uppermost sides of the bases **13A** and **13B** configure an opening **16B** of the housing body **16**. The opening **16B** functions as an insertion part of the housing body **16** for the module **3**. In accordance with this, as illustrated in FIG. 3, the module **3** can be easily inserted into the housing body **16** from the opening **16B**.

(38) In addition, the insertion part of the housing body **16** for the module **3** may be disposed in a part of the housing body **16** but is not limited to the uppermost side. In other words, a side on which the seam allowance **133** is not disposed among edges of the first base part **131** may be any one side in the x-axis direction. As another example, as illustrated in FIG. 4, lateral sides of the

bases **13A** and **13B** may configure an opening **16C** of the housing body **16**, or both lateral sides may configure an opening **16C**. Also in this case, as illustrated in FIG. **4**, the module **3** can be easily inserted into the housing body **16** from the opening **16C**.

(39) In the module **3**, the connector electrodes **31** and **32** are disposed on both faces including one face and a rear face side of the face. As one example, the module **3** may be a rectangular parallelepiped, and the connector electrodes **31** and **32** are disposed on two faces facing each other among six faces thereof. The connector electrodes **31** and **32** respectively function as a positive electrode and a negative electrode.

(40) As illustrated in FIG. **3** or FIG. **4**, the module **3** is housed in the housing body **16** such that the connector electrode **31**, which is a positive electrode, faces the connector electrode **14** which is a positive electrode, and the connector electrode **32**, which is a negative electrode, faces the connector electrode **15** which is a negative electrode. The connector electrodes **14** and **15** are sewed into the first base part **131** such that they are exposed to an inner face of the housing body **16**. For this reason, the connector electrodes **31** and **32** are brought into contact with the connector electrodes **14** and **15** exposed to the inner face of the housing body **16**. In accordance with this, the connector electrodes **14** and **15** are respectively connected to the connector electrodes **31** and **32** of the module **3**. As a result, the electrodes **11** and **12** are respectively connected to the connector electrodes **31** and **32** of the module **3** through the connector electrodes **14** and **15** connected to ends thereof.

(41) Since the connector electrodes **14** and **15** are sewn such that they are exposed to the inner face of the housing body **16**, only by housing the module **3** in the housing body **16**, the connector electrodes **14** and **15** can be easily connected to the connector electrodes **31** and **32**. At this time, since the widths of the connector electrodes **14** and **15** are larger than the widths of the electrodes **11** and **12**, positioning with the connector electrodes **31** and **32** becomes unnecessary or easy. In addition, since the module **3** is not exposed to the outside, damage and contamination can be prevented, and damage of connection to the connector electrodes **14** and **15** can be also prevented.

(42) Preferably, at least one of the sensor **1** and the module **3** further includes a contact part which brings the connector electrodes **31** and **32** of the module **3** housed in the housing body **16** and the connector electrodes **14** and **15** of the inside of the housing body **16** to be in contact with each other.

(43) FIG. **5** is a schematic diagram illustrating a specific example of the contact part. The contact part of the sensor **1** includes a pressing member which presses a module **3** housed in the housing body **16** to the inner face of the housing body **16**. For example, the pressing member may be a belt **17** which is wound around the housing body **16** from the outer side. Preferably, the belt **17** is composed of an elastic material such as rubber and fixes the module **3** after housing from the outside of the housing body **16**. Other than a member of an opening-rotation type such as the belt **17**, the pressing member may be a member of an interposition type of a clip shape which interposes the housing body **16** therebetween from the outside, a member such as a button or a hook and loop fastener disposed inside of the housing body **16** which closes an opening **16B** of the housing body **16** or may be a combination thereof.

(44) The contact part of the module **3** is a protrusion disposed on the front face of the module **3**. For example, the protrusion may be a convex part **33** disposed on a face on which the connector electrodes **31** and **32** are installed. In addition, both the sensor **1** and the module **3** may have such structures. By including a contact part in at least one of the sensor **1** and the module **3**, the connector electrodes **31** and **32** of the module **3** housed in the housing body **16** and the connector electrodes **14** and **15** of the inside of the housing body **16** are brought into contact with each other. For this reason, connection of these connector electrodes is reliably formed. In addition, the connection is maintained.

(45) FIG. **6** is a schematic diagram illustrating a configuration of the module **3**. The module **3** is a sensor module, detects power generation of the electrodes **11** and **12** included in the sensor **1**, and

outputs a signal representing a detection result. In more details, as illustrated in FIG. 6, the module 3 includes a capacitor 34. The capacitor 34 is connected to the sensor 1 through the connector electrodes 31 and 32 and the connector electrodes 14 and 15 and saves electric power generated using the electrodes 11 and 12. In FIG. 6, the connector electrodes 31 and 32 and the connector electrodes 14 and 15 are omitted.

(46) The module 3 includes an intermittent power converting circuit 35. The capacitor 34 is connected to a power supply terminal of the intermittent power converting circuit 35. The intermittent power converting circuit 35 causes the capacitor 34 to operate as an operation power supply. The intermittent power converting circuit 35 monitors a charge voltage of the capacitor 34.

(47) A radio transmitter 36 is connected to the intermittent power converting circuit 35. The radio transmitter 36 outputs a communication signal using radio communication. For example, the radio communication is Bluetooth (a registered trademark), Bluetooth Low Energy (a registered trademark), or the like.

(48) The intermittent power converting circuit 35 detects whether the charge voltage of the capacitor 34 has reached a setting voltage set as a drive condition of the radio transmitter 36. When the reach has been detected, the intermittent power converting circuit 35 supplies electric power charged in the capacitor 34 to the radio transmitter 36. In accordance with this, a radio signal SG is output from the radio transmitter 36.

(49) In accordance with the intermittent power converting circuit 35 supplying electric power to the radio transmitter 36, the electric power of the capacitor 34 is consumed. When the electric power is consumed, the electric potential of the capacitor 34 is lowered, and the intermittent power converting circuit 35 stops its operation. In accordance with this, supply of the electric power to the radio transmitter 36 stops. When the electrodes 11 and 12 are generating power, the capacitor 34 is charged again.

(50) When the electrodes 11 and 12 are generating power, the capacitor 34 repeats charging and discharging. In accordance with this, output of a radio signal SG from the radio transmitter 36 becomes intermittent. Thus, the radio signal SG output from the radio transmitter 36 becomes a detection signal representing detection of a liquid present between the electrodes 11 and 12.

(51) An output interval H of radio signals SG from the radio transmitter 36 depends on a charging speed of the capacitor 34. The larger the amount of generated power, the higher the charging speed. Thus, the larger the amount of generated power of the electrodes 11 and 12, the shorter the output interval H of radio signals SG from the radio transmitter 36. In addition, the smaller the amount of generated power, the shorter the output interval of radio signals.

(52) The sensor 1 is disposed in a member mounted in a human body as an example and is used for detection of presence/absence of a liquid in the member. For example, the member mounted in the human body is a diaper. By disposing the sensor 1 in a member mounted in a human body, the housing body 16 is also disposed in the member. For this reason, the module 3 can be inserted into the member mounted in a human body. In accordance with this, presence/absence of a liquid in a member mounted in a human body can be detected using a wearable device.

(53) FIG. 7 is a schematic diagram illustrating an overview of a case in which the sensor 1 is disposed in the absorption member main body 5 as an example of a member mounted in a human body and is used in a detection system 100 detecting presence/absence of a liquid in the absorption member main body 5. The detection system 100 detects a liquid present in the absorption member main body 5 using the sensor 1 installed in the absorption member main body 5. FIG. 8 is a cross-sectional view of the absorption member main body 5 taken along line VIII-VIII illustrated in FIG. 7. FIG. 9 is a cross-sectional view of the absorption member main body 5 taken along line IX-IX illustrated in FIG. 7. The absorption member main body 5 has a basic configuration as a diaper as an example.

(54) The absorption member main body 5 has an almost rectangular shape which is long in one direction in the plan view in an expanded state. In the absorption member main body 5, a length

direction will be defined as a Y axis, a lateral direction orthogonal to the length direction will be defined as an x axis, and a direction orthogonal to those will be defined as a Z axis. The Y-axis direction coincides with a front-back direction at the time of mounting the absorption member main body in a mounting person. In other words, a position having a large positive value among positions in the Y-axis direction will be referred to as a front side, and a position having a large negative value will be referred to as a rear side. The X-axis direction coincides with the width direction. The Z-axis direction coincides with a vertical direction at the time of mounting, a positive direction of the Z axis is an upward direction and a side toward a mounting person at the time of mounting the absorption member main body in the mounting person, and a negative direction of the Z axis is a downward direction and a side toward an outer side at the time of downwardly mounting the absorption member main body in the mounting person.

(55) The absorption member main body **5** includes a front sheet **51**, a rear sheet **52**, and an absorption body **53**. The absorption body **53** is disposed between the front sheet **51** and the rear sheet **52**. A front face of the absorption member main body **5** represents a face of a side brought into contact with a mounting person when the absorption member main body **5** is mounted in the mounting person. Thus, excrement of the mounting person is given from the front sheet **51** side to the absorption body **53**.

(56) The front sheet **51** is a liquid transmissive sheet having an approximately rectangular shape. For example, the front sheet **51** is formed using a nonwoven fabric or a woven fabric. The front sheet **51** is brought into contact with a skin of a mounting person when the absorption member main body is mounted in the mounting person. The front sheet **51** is configured such that a transmitted liquid is difficult to turn back to the mounting person side by improving transmissivity of the liquid. In accordance with this, urine excreted from the mounting person can be quickly transmitted to the absorption body **53**. Thus, even when there is excretion, as long as there is remaining power in an absorption capability of the absorption body **53**, the front sheet **51** does not substantially hold urine and is in a substantially dry state. As a result, urine is inhibited from being brought into contact with the skin of the mounting person.

(57) The rear sheet **52** is a non-liquid transmissive sheet having an approximately rectangular shape. The rear sheet **52** is formed using a water-proof member having a water-proof film or the like. The rear sheet **52** prevents urine absorbed by the absorption body **53** from leaking to the outside.

(58) The absorption body **53** is composed of an absorptive fiber such as a pulp and a high-absorbent polymer. In accordance with the high-absorbent polymer, much liquid can be held by the absorption body **53**. The absorption body **53** is a mat body having an approximately rectangular shape which is long in one direction. The absorption body **53** has a longitudinal direction which approximately coincides with the Y-axis direction of the absorption member main body **5** and is disposed over a front side and a rear side of the absorption member main body **5**. Thus, together with absorbing urine excreted on the front side, the absorption body **53** can absorb urine exceeding the amount of absorption of the front side on the rear side.

(59) The sensor **1** is disposed between the absorption body **53** and the rear sheet **52**. In other words, the sensor **1** is disposed on a rear sheet **52** side of the absorption body **53**. The sensor **1** functions as a urine power generation battery.

(60) The sensor **1** is disposed in the absorption member main body **5** with positive/negative directions of the Y-axis direction approximately coinciding with positive/negative directions of the Y-axis direction of the absorption member main body **5**. Thus, the electrodes **11** and **12** are disposed along the Y-axis direction apart from the absorption member main body **5** in the X-axis direction.

(61) FIG. **8** illustrates a cross-section of the sensor **1** on the rear side of the absorption member main body **5** in which the sensor **1** is disposed, and FIG. **9** illustrates a cross-section of the sensor **1** at the position of the housing body **16** on the front side. Referring to FIG. **8**, on the rear side of the

absorption member main body 5, the electrodes 11 and 12 are disposed apart from each other in the X-axis direction. Referring to FIG. 9, on the rear side of the absorption member main body in a space 16A between bases 13A and 13B composing the housing body 16, the module 3 is housed such that the connector electrodes 31 and 32 are respectively brought into contact with the connector electrodes 14 and 15 disposed in the housing body 16.

(62) The sensor 1 is disposed in the absorption member main body 5 such that the bases 13A and 13B overlapping each other are brought into contact with the rear sheet 52 side of the absorption body 53.

(63) Since the bases 13A and 13B overlapping each other are disposed with being brought into contact with the absorption body 53, a liquid absorbed by the absorption body 53 moves to the bases 13A and 13B overlapping each other. The bases 13A and 13B which have absorbed a liquid from the absorption body 53 overlapping each other become a path for generation of a current between the electrodes 11 and 12. For this reason, in accordance with the electrodes 11 and 12 being brought into contact with a liquid which has been absorbed by the absorption body 53, a current is generated. In accordance with this, the sensor 1 is used for detection of liquid absorption of the absorption body 53.

(64) The larger an area in which the electrodes 11 and 12 are brought into contact with a liquid, the larger the amount of generated power. In other words, the electrodes 11 and 12 are disposed to be brought into contact with the absorption body 53 with the longitudinal direction approximately coinciding with the longitudinal direction of the absorption body 53, and thus the more amount of a liquid absorbed in the absorption body 53, the larger the amount of generated power. Thus, as illustrated in FIG. 6, the output interval H of radio signals SG from the radio transmitter 36 represents an amount of liquid absorbed by the absorption body 53.

(65) As illustrated in FIG. 7, the detection system 100 may include a receiver 7. The receiver 7 receives a radio signal SG transmitted from the radio transmitter 36. The receiver 7 may be connected to a management device 9. Reception data is given to the management device 9 from the receiver 7. The management device 9 is a computer including a processor 91 and a memory 92 and, as an example, may be a terminal device such as a smartphone.

(66) By executing a program stored in the memory 92, the processor 91 can perform a process relating to a degree of absorption of the absorption body 53 of the absorption member main body 5. For example, the process relating to the degree of absorption of the absorption body 53 of the absorption member main body 5 is a process of determining necessity/unnecessity of replacement of the diaper. When the liquid reaches the rear side of the absorption body 53, the absorption member main body 5 determines a replacement time. In a case in which it is determined that replacement of the diaper is necessary, the output device 93 of the management device 9 outputs necessity of the replacement of the diaper.

(67) The inventors and the like performed a test for evaluating sensing of the sensor 1 according to this embodiment. In the test, after the sensor 1 was disposed in the absorption member main body 5, and 600 ml of liquid is injected into the absorption member main body 5, presence/absence of reception of a radio signal SG in the receiver 7 was measured, and measurement results illustrated in FIG. 10 were acquired. A vertical axis represents presence/absence of reception of a radio signal SG, "0" represents absence of reception, and "1" represents presence of reception. A horizontal axis represents an elapse time after injection of 600 ml of liquid into the absorption member main body 5. Thus, the measurement results illustrated in FIG. 10 represent timings at which a radio signal SG is received after injection of illiquid using an elapse time from injection of the liquid.

(68) From the results illustrated in FIG. 10, reception of a radio signal SG starts at a timing at which 600 seconds elapses after injection of liquid. In other words, it can be understood that, in a case in which 600 ml of liquid is injected into the absorption member main body 5, by using the sensor 1, the liquid is detected after 600 seconds.

(69) The present invention is not limited to the embodiment described above, and various

modifications can be made. In the embodiment described above, although urine has been described as an example of liquid, the liquid is not limited to the urine, and various kinds of liquid can be detected as long as the liquid generates power in the electrodes by being brought into contact with the electrodes. The sensor can be used as a water leakage sensor or a rainwater sensor.

Claims

1. A power generation sensor comprising: a housing body in which a module receiving generated power is housed; a module with a first connector electrode disposed on a front face; a second connector electrode disposed on an inner face of the housing body; and a power generation electrode further connected to the second connector electrode, wherein the first connector electrode and the second connector electrode are connected to each other.
 2. The power generation sensor according to claim 1, further comprising a contact part bringing the first connector electrode of the module housed in the housing body and the second connector electrode into contact with each other.
 3. The power generation sensor according to claim 2, wherein the contact part includes a pressing member pressing the module housed in the housing body to the inner face of the housing body.
 4. The power generation sensor according to claim 1, wherein the housing body includes an insertion part for the module.
 5. The power generation sensor according to claim 1, wherein a positive electrode of the first connector electrode is disposed on a first front face of the module, and a negative electrode of the first connector electrode is disposed on a face opposite the first front face, wherein a positive electrode of the second connector electrode is disposed on a first face of the inner face of the housing body, and a negative electrode of the second connector electrode is disposed on a second face positioned in a first direction with respect to the first face, wherein the power generation electrode includes a positive electrode connected to the positive electrode of the second connector electrode and a negative electrode connected to the negative electrode of the second connector electrode, and wherein the positive electrode and the negative electrode of the power generation electrode are disposed apart from each other in a second direction different from the first direction.
-